CRISISFORGE

Decision-Focused Market Simulation under Regime Shifts

Factor-Structured Temporal Diffusion, Asset-Level Tail Risk, and Robust Portfolio Control

Technical research report
Public-core validation release v0.3.0
28 July 2026

PUBLIC-CORE VALIDATION

SEALED TEST

NEGATIVE RESULTS RETAINED

Contents

This report separates implemented evidence, engineering pilots, and unidentified extensions. Lower distribution and risk scores are better unless explicitly stated.

Technical summary

CrisisForge is a reproducible research system for testing whether structured and generative financial scenarios improve multi-asset tail-risk measurement and portfolio decisions. The implemented system combines a train-only factor representation, a sticky Gaussian hidden Markov model, regime-weighted factor dynamics, a stochastic factor-to-asset map, a one-shot temporal diffusion pilot, asset-level VaR/Expected Shortfall/co-crash metrics, empirical CVaR, a tractable 1-Wasserstein robust-CVaR decision layer, and a separate semi-synthetic structural counterfactual experiment.

The main result is not that generative AI wins. Strong conventional models remain hard to beat. On 74 common validation origins, the Stage 1 switching-factor model has worse energy and variogram scores than leading historical and block-bootstrap baselines. Its joint VaR-ES point score is close to the best conventional result, but every paired joint VaR-ES interval includes zero. The Stage 2 diffusion run is an engineering pilot with four reporting origins and cannot establish superiority. Historical empirical CVaR has lower realized validation ES, drawdown, turnover, and transaction cost than Stage 1 scenario CVaR; the registered Wasserstein radii barely change allocations. A known-SCM counterfactual engine recovers oracle paths to numerical precision but deteriorates substantially under structural misspecification.

These findings matter because they reject a common implicit premise: statistical or architectural sophistication does not automatically improve risk decisions. The public-core contribution is an auditable framework, a sealed test contract, and a retained set of negative results that define the next valid experiments.

No reported result uses the post-2019 test set. The study does not establish investability, a live-trading edge, real-market causal identification, or an AI system capable of predicting the next crisis.

The implemented system is broad, but its evidence levels differ

The core path moves from governed public data to latent factors and states, then to asset scenarios, risk measures, and decisions. Conventional generators provide the comparison set. Tail weighting is integrated only in the small diffusion pilot. The causal track is deliberately separate because it relies on known semi-synthetic structural equations.

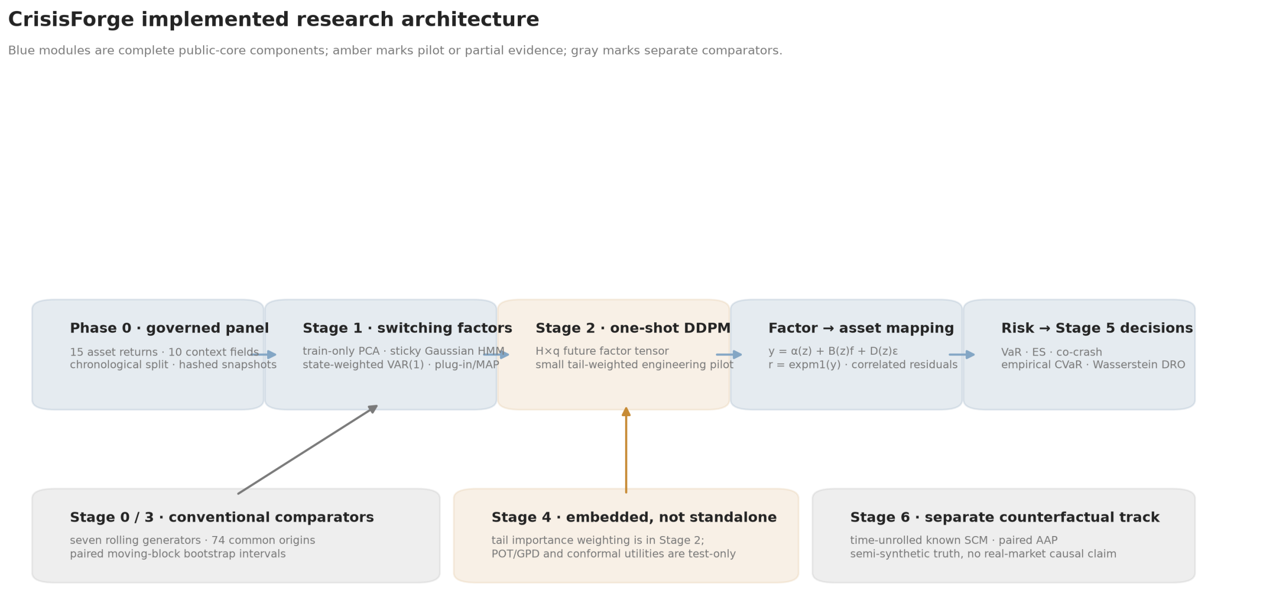

Figure 1. *Implemented CrisisForge research architecture*

The diagram should be read as an evidence map. Blue boxes are completed public-core components; amber identifies pilot or partial evidence; gray identifies comparators or a separate semi-synthetic validation track. “Switching factors” means a multi-stage PCA-HMM-VAR estimator with plug-in parameters. It is not a joint Bayesian posterior. The factor-to-asset equation is genuinely implemented, including correlated idiosyncratic residuals, but its parameters are fixed after training rather than drawn from a posterior.

The research frontier supports the components, not a presumption of success

The design is anchored in several distinct primary literatures. Hamilton's Markov-switching model and Kim's switching state-space filtering establish latent-state and filtering foundations. Recent peer-reviewed work by Urga and Wang and Barigozzi and Massacci develops high-dimensional or large-panel regime-switching factor methods.

For one-shot generation, TimeDiff provides an archival precedent for non-autoregressive multi-step diffusion, while FTS-Diffusion addresses financial time-series structure. Finance-specific factor diffusion remains frontier evidence: the closest direct factor-structured paper, Chen et al., is a preprint and is used only to motivate a falsifiable hypothesis. Tail-focused generation has a peer-reviewed comparator in Tail-GAN; conditional EVT is grounded in McNeil and Frey.

The decision layer follows the tractability results for Wasserstein DRO and the distinction between prediction quality and downstream decision loss in decision-focused learning. The counterfactual track is aligned with accepted time-series intervention work such as DoFlow and DiffCATS, but those sources do not identify monetary-policy effects in observational financial data.

The full status ledger, publication metadata, and claim-to-source limits are in `docs/literature_review.md`. Every 2025-2026 manuscript without an archival record is labeled as a preprint or working paper.

Data governance prevents the test set from becoming a hidden tuning resource

The public matrix has 6,465 complete common-endpoint observations, 15 asset-return targets, and ten context variables. The targets comprise 12 value-weighted Kenneth French industry research portfolios and three duration-convexity Treasury return proxies derived from official par yields. The context contains Treasury yield levels and slope, New York Fed EFFR, and five backward-looking industry-return features.

The assets are method-development proxies. French industry portfolios are retrospective research constructs, not ETFs. Treasury targets are approximations, not observed total-return indices. The panel is suitable for falsifying scenario methods; it is not sufficient for tradeability or execution claims.

Property	Registered public core
Complete model rows	6,465
Asset targets	15
Context features	10
First model date	5 July 2000
Last model date	29 May 2026
Phase 0 quality gates	23/23 passed
Formal warning	French source ends 60 days before requested endpoint

The split is chronological: 3,367 training rows through 2013, 1,499 validation rows through 2019, and 1,599 sealed test rows from 2020. Experiment runners scan the processed Parquet matrix only through the registered validation boundary.

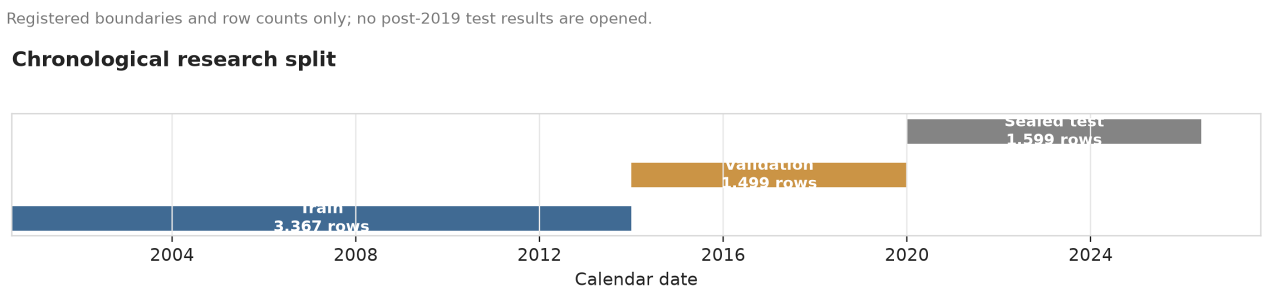

Figure 2. *Chronological train, validation, and sealed-test spans*

The gray bar is a governance commitment rather than reported evidence. Its dates and row count are registered, but no post-2019 model score or portfolio result appears in this report. Phase 0 constructs and quality-checks those rows; “sealed” means they are governed-excluded from estimator fitting, checkpoint selection, model scoring, and portfolio decisions. Using that span for model evaluation requires a new signed release decision; it cannot be used to improve the current model family and still be called a test.

The Phase 0 manifest contains 106 governed file records and has SHA-256 `f051ec35236a481858b67c5b1e7136f1698036832f427efba521dbf3fcd36d70`. Stages 0-5 bind that exact snapshot and clean experimental commit `b6891133bdb6b96e1e23c6bea3bd033ea9685c7c`; Stage 6 v2 is bound to its separately recorded clean implementation commit `23a4c9ad20683e5e94f7154472073a96d18f90be`.

Conventional time-dependence baselines lead the structured model

Stage 0 evaluates seven conventional generators on 74 non-overlapping H=20 validation origins, using 1,000 scenarios per origin. The model set includes historical IID, a moving-block bootstrap, filtered historical simulation with EWMA volatility, Gaussian shrinkage, multivariate Student-t, a Student-t copula, and VAR residual bootstrap.

Stage 1 estimates four PCA factors, a four-state sticky Gaussian HMM, weighted state-specific VAR(1) dynamics, and a regime-weighted Gaussian observation map. The four factors explain 85.995% of standardized training log-return observation-space variance. The HMM converges after 37 iterations from the selected one of 12 initializations. Its smallest state has 5.23% soft occupancy, above the registered 3% diagnostic warning threshold; all state-specific VAR dynamics are stable. The separate 0.1% HMM minimum-state-weight setting is an estimation safeguard, not a reported occupancy target.

Energy and variogram scores use the 15-asset geometrically compounded H=20 return vector. VaR, ES, joint VaR-ES, and violation rates use a frozen equal-weight portfolio with w_i=1/15.

Model	Energy ↓	Variogram ↓	Joint VaR-ES ↓	VaR violation rate
Filtered historical EWMA	**0.089739**	0.542024	-0.947758	2.70%
Moving block	0.089811	**0.532226**	-0.949032	**5.41%**
VAR residual bootstrap	0.091236	0.563301	-0.948535	2.70%
Student-t copula	0.091478	0.584204	-0.947724	2.70%
Student-t elliptical	0.091630	0.578555	-0.949064	4.05%
Historical IID	0.091732	0.571600	-0.946065	2.70%
Gaussian shrinkage	0.091803	0.582730	-0.948546	4.05%
Stage 1 switching factor	0.094315	0.588639	**-0.949124**	1.35%

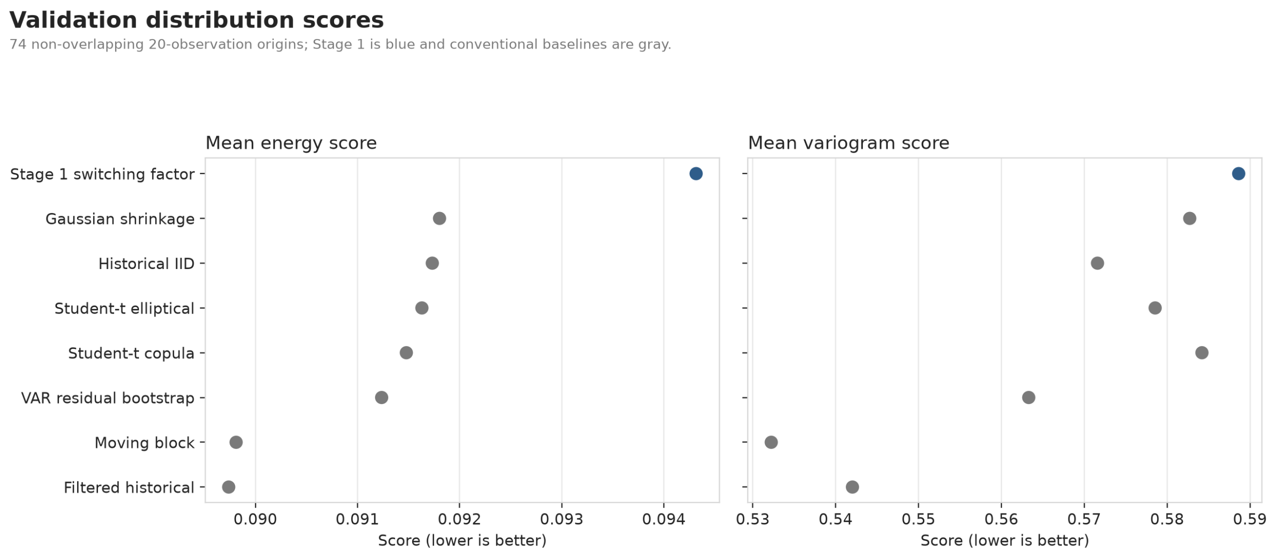

Figure 3. *Validation energy and variogram scores*

Lower scores are better. Stage 1 is visibly behind every conventional model on energy score and behind the leading block/historical methods on variogram score. Its joint VaR-ES point estimate is marginally lower than the best Stage 0 value, but the difference is 0.000060, too small to interpret without paired uncertainty. Its one VaR breach in 74 blocks indicates a conservative forecast; it does not establish improved calibration. Bold marks the lowest point estimate in each score column; in the VaR-violation column it marks the rate closest to the nominal 5%, not the numerically smallest rate.

No baseline wins every metric. Filtered historical simulation has the lowest energy score; moving blocks have the lowest variogram score, which emphasizes pairwise return-difference structure, and give the closest 5% VaR coverage; and the Student-t elliptical model has the leading Stage 0 joint tail score. The dispersion of winners is itself evidence against choosing a model from one aggregate metric.

Paired uncertainty shows no reliable Stage 1 advantage

Stage 3 compares Stage 1 and each Stage 0 generator at identical origins. A circular moving-block bootstrap with block length four and 10,000 draws preserves local dependence among ordered origin differences. Intervals are exploratory and not multiplicity-adjusted.

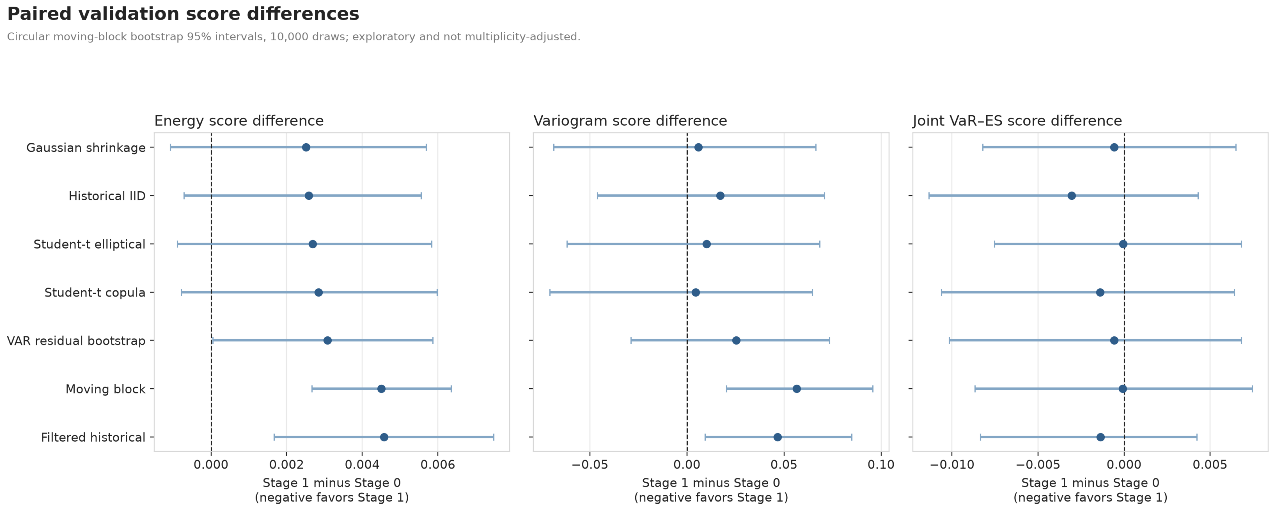

Figure 4. *Paired Stage 1 minus Stage 0 score intervals*

For energy score, the Stage 1 difference is +0.00458 versus filtered historical simulation and +0.00450 versus moving block; both 95% intervals exclude zero. For variogram score, the differences are +0.0466 and +0.0564, again with intervals above zero. Positive differences favor Stage 0. All seven joint VaR-ES intervals cross zero.

Twenty-one intervals are reported across three metrics and seven baselines. None clearly favors Stage 1; five exclude zero in the direction unfavorable to Stage 1. This is descriptive inferential evidence, not a causal treatment effect of model class. Stage 0 refits a rolling 1,500-row window at every origin, whereas Stage 1 freezes parameters at the end of training and updates only filtered state beliefs. The comparison therefore mixes architecture and update policy.

The latent states are statistically distinct, not automatically economic labels

The four HMM states differ in mean, volatility, persistence, and occupancy. State 0 has the lowest weighted market mean and highest weighted volatility; state 3 has the highest mean, lowest volatility, and largest occupancy. State 0 also receives high ex-post probability during parts of 2008. These observations support an ex-post "high-volatility/negative-mean profile," but do not justify naming the component a structural crisis regime.

Figure 5. *Training-only latent state profiles*

The distinction is important. State numbers are statistical labels chosen for reproducibility, not observed market types. The current HMM has Gaussian factor emissions and homogeneous transitions. It does not contain macro-dependent transition probabilities or a Bayesian parameter posterior. Treating a state as an economic cause would exceed what the estimator identifies.

The factor-to-asset map closes the financial modeling loop

The scenario generator operates in four-dimensional factor space, while VaR, ES, co-crash, and portfolio optimization require 15 asset-level simple returns. CrisisForge therefore estimates a state-specific observation equation in log-return space:

$$y\_t = log(1 + r\_t) = alpha\_z\_t + B\_z\_t\ f\_t + D\_z\_t\ epsilon\_t$$
$$Cov(D\_z\ epsilon\_t) = D\_z\ R\_z\ D\_z$$
$$r\_t = expm1(y\_t)$$

In the implementation, each state uses a weighted ridge map and a shrunk Gaussian residual covariance. Scenario generation draws a complete future state path, applies the corresponding B_z, adds a correlated residual draw in y-space, and converts the result back to simple returns with expm1. It never probability-averages state-specific loading or Cholesky matrices. This retains modeled asset-specific residual variation and residual dependence, avoiding the false conclusion that every asset is a deterministic function of four common factors.

This layer is structurally necessary but not yet fully ablated. A fixed-map comparator, Student-t residuals, and parameter-posterior propagation remain future experiments.

The diffusion run validates engineering, not model superiority

Stage 2 learns a complete 20\times4 future factor tensor in one reverse diffusion process. Context consists of a 60-observation factor/macro history and the origin's filtered state probabilities. The public pilot uses 658 train windows, 37 tuning windows, 12 diffusion steps, 64 scenarios per origin, three base epochs, two tail-weighted fine-tuning epochs, and only four late-validation reporting origins.

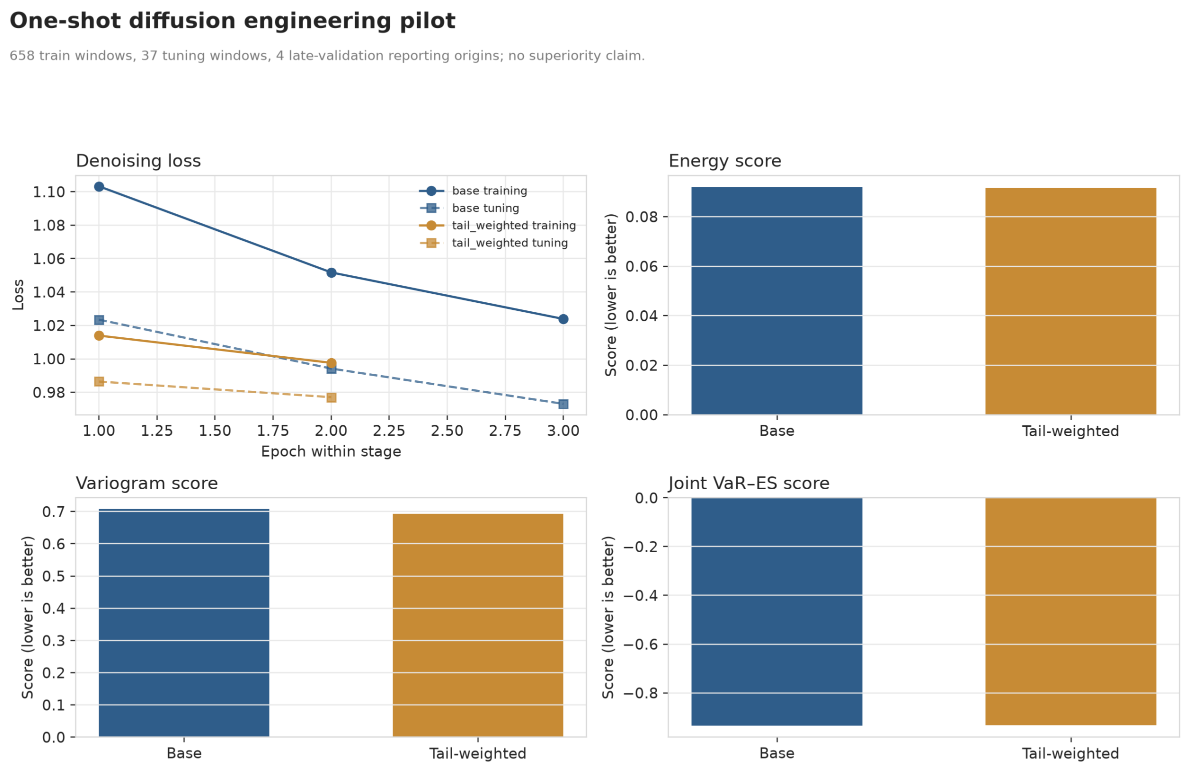

Figure 6. *One-shot diffusion pilot training and descriptive scores*

Pilot variant	Energy ↓	Variogram ↓	Joint VaR-ES ↓	VaR breaches
Base DDPM	0.092049	0.707209	**-0.934902**	0/4
Tail-weighted DDPM	**0.091680**	**0.693524**	-0.933368	0/4

Tail weighting improves energy by 0.40% and variogram by 1.94%, but worsens the joint VaR-ES score by 0.00153. This is a metric trade-off between multivariate distribution scores and the joint VaR-ES score, not evidence of calibrated tail improvement. Four origins cannot support confidence intervals, coverage claims, or a ranking against 74-origin models. Future HMM paths are also drawn independently of generated factor paths conditional on the same origin belief, so their joint dynamic consistency is not guaranteed.

The valid conclusion is narrow: the Python implementation can form chronological windows, standardize train-only inputs, train and checkpoint a one-shot DDPM, perform a tail-weighted fine-tune, generate factor tensors, reconstruct asset-level scenarios, and retain a complete audit trail. POT/GPD and rolling conformal utilities exist and are tested but were not integrated into this run.

Co-crash validation is event-limited

Co-crash probability is defined from training-only asset thresholds and the fraction of assets simultaneously breaching them over an H=20 block. At every evaluated origin-74 Stage 0/Stage 1 origins and four Stage 2 origins-the realized validation co-crash label is zero; generated scenario sets may still assign nonzero co-crash probabilities.

This means the Brier scores mainly reward forecasting probabilities near zero. They do not establish the ability to distinguish an actual co-crash. A meaningful study needs either more evaluation origins, a less extreme preregistered event definition, a longer licensed history, or a separate stress-evaluation design. The event definition cannot be changed after inspecting results and still be treated as confirmatory.

Complex scenarios did not produce lower observed decision risk

Stage 5 converts scenario paths to geometrically compounded H=20 asset returns, then solves long-only, fully invested 95% CVaR problems with a 20% asset cap, 40% L1 turnover cap, and 10-basis-point linear transaction-cost rate. Four Wasserstein radii are reported as separate sensitivities, not tuned winners.

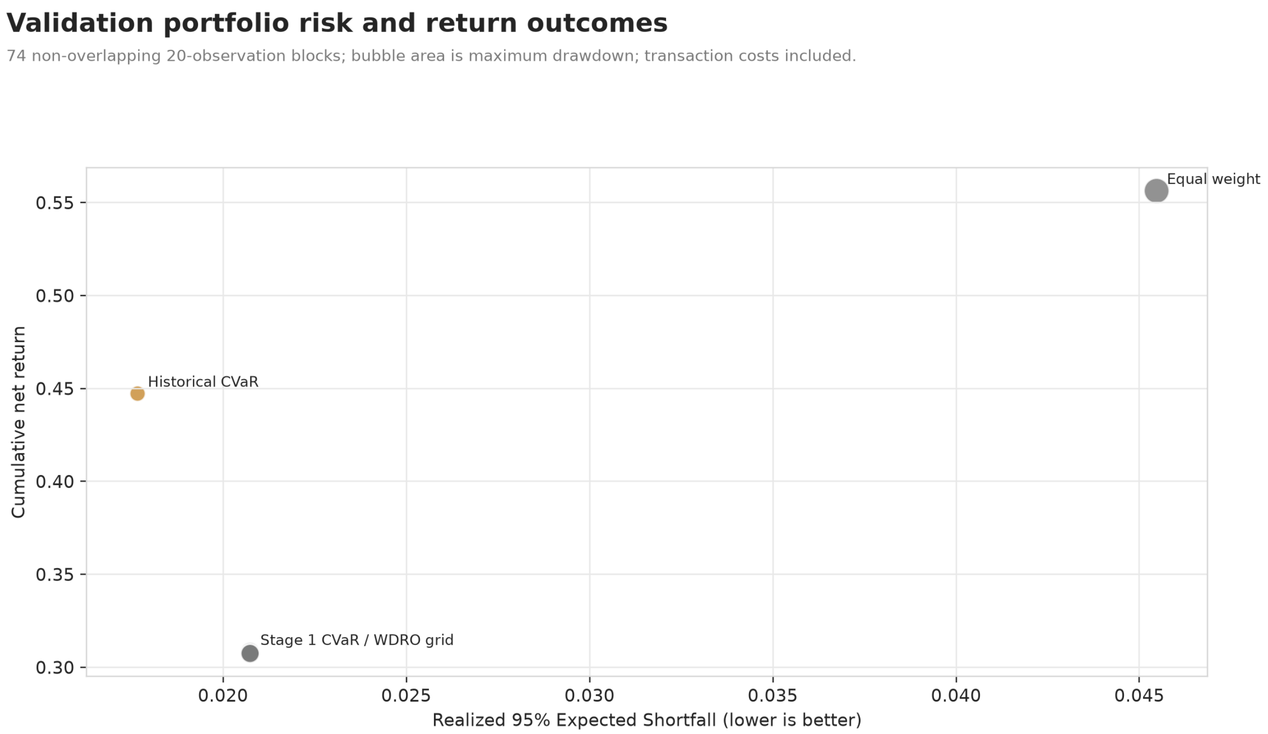

Figure 7. *Validation portfolio Expected Shortfall and return outcomes*

Decision rule	Cumulative net return	Realized ES ↓	Maximum drawdown ↓	Mean L1 turnover
Equal weight	**0.5564**	0.04547	0.08769	**0.01956**
Historical empirical CVaR	0.4473	**0.01767**	**0.03411**	0.04724
Stage 1 scenario CVaR	0.3083	0.02074	0.04782	0.16885
Stage 1 WDRO grid	0.3074-0.3083	0.02074	0.04782	0.16863-0.16885

Equal weight has the highest net return but the weakest risk protection. Historical CVaR has the lowest observed validation ES and drawdown among the tested rules. Relative to Stage 1 scenario CVaR, it also has 13.9 percentage points more cumulative net return and much lower turnover. Stage 1 produces a lower realized VaR, but its ES is higher, indicating larger observed conditional tail losses beyond the threshold. These rankings are descriptive: they use 74 non-overlapping block outcomes, and no paired sampling interval or multiplicity-adjusted test was computed.

The WDRO radii 0.0001 and 0.00025 return the same solution as empirical Stage 1 CVaR; 0.0005 and 0.001 produce only tiny changes and no ES or drawdown improvement. One plausible explanation is that the 20% position cap already constrains the l_infinity norm penalized by the L1-ground-cost Wasserstein dual. This explanation is interpretive rather than separately identified. What the evidence establishes is a locally flat validation sensitivity, not a robust-decision advantage.

The counterfactual engine is correct under truth and fragile under misspecification

Stage 6 uses a known, time-unrolled, regime-switching SCM with within-period order policy → yield → liquidity → credit → equity → volatility and lagged equity-to-policy feedback. A five-step policy intervention is applied to 5,000 paired H=20 paths with shared exogenous noise.

Abduction-action-prediction recovers the oracle counterfactual with maximum error 4.08\times10^-17. This verifies the implementation in a model where every structural equation and innovation is known. Two deliberately incorrect models then alter the yield-transmission coefficient or remove lagged equity feedback.

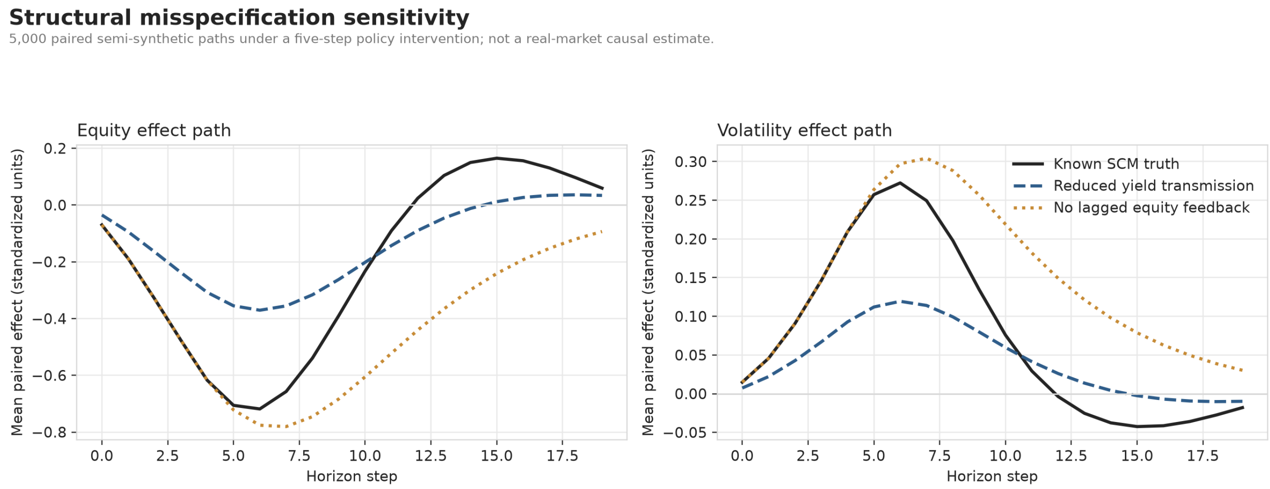

Figure 8. *Counterfactual effect-path sensitivity*

Misspecification	Equity path RMSE	Equity tail-effect error	Volatility path RMSE	Volatility tail-effect error
Reduced yield transmission	0.1885	0.6019	0.0731	0.5466
No lagged equity feedback	0.2782	3.2142	0.0931	3.4354

For outcome j, Stage 6 defines path loss as L_j = s_j x sum(h = 1..20) X_(j,h), with s_equity=-1 and s_volatility=+1. The tail effect is ES_0.95(L_j^treated) -ES_0.95(L_j^reference). Lower cumulative equity outcomes are therefore losses, while higher cumulative volatility is a stress loss. “Tail-effect error” is the absolute error in that 20-step path-loss ES effect. Path RMSE compares the estimated and known mean intervention-effect paths.

The misspecified path RMSE range, 0.0731-0.2782 standardized units, shows that counterfactual outputs depend materially on structural assumptions even when the numerical AAP machinery is correct. The controlled direct effect is zero because the registered mediator schedule blocks all policy-to-outcome paths by design.

Nothing in this experiment identifies the effect of a Federal Reserve action in observed markets. A real-data causal claim would require a separately registered treatment, identification strategy, support analysis, and external validity argument.

What the experiments establish-and what they do not

Established on the validation evidence

- The public data pipeline is reproducible, availability-aware, and bound to a fixed content-addressed snapshot.
- The switching-factor and stochastic observation pipeline produces finite, schema-valid asset-level scenarios with stable fitted state-specific dynamics.
- Strong historical and block-bootstrap methods lead the structured model on several joint-distribution metrics.
- The one-shot diffusion path runs end to end and displays a visible trade-off between multivariate distribution scores and the joint VaR-ES score.
- Generator complexity does not automatically improve realized CVaR decisions.
- The known-SCM counterfactual implementation is numerically correct and structurally sensitive.
- Negative results, failures, configurations, and receipts are retained.

Not established

- Any superiority on the sealed test set.
- Full Bayesian posterior uncertainty or parameter propagation.
- Tail calibration from EVT, conformal methods, or ESR evidence.
- Co-crash discrimination.
- A selected Wasserstein radius or robust-decision advantage.
- Investability, implementability, or live-trading profitability.
- A real-market causal effect or a model that predicts unseen crises.

Robustness, uncertainty, and remaining gaps

The most important robustness result is the paired Stage 3 comparison. It directly tests origin-level score differences and rejects a narrative of broad Stage 1 superiority. The most important decision robustness result is the near-flat WDRO radius grid. The most important causal robustness result is the deterioration under two known structural misspecifications.

Important unrun diagnostics remain:

- direct-asset, unconditional, hard-state, and fixed-mapping diffusion ablations;
- rolling versus frozen estimation-policy controls;
- Student-t observation residuals;
- sliced Wasserstein distance, MMD, return/squared-return ACF, leverage, and lower-tail dependence diagnostics;
- Monte Carlo intervals for risk measures;
- integrated POT/GPD, sequential conformal VaR, and ESR testing;
- validation-calibrated Wasserstein radius selection; and

- posterior-versus-plug-in uncertainty propagation.

The public factor representation also uses PCA scores rather than economically identified factors. This improves statistical compression but makes factor names rotation-dependent. The homogeneous Gaussian HMM may conflate volatility, location, and persistence. The validation contains only 74 non-overlapping blocks, and crisis-like events remain rare.

Recommended next experiments

1. **Run a confirmatory Stage 2 evaluation on all 74 validation origins.** Freeze the current architecture and add direct, unconditional, hard-state, and fixed-map ablations before expanding capacity.
2. **Equalize update policies.** Compare rolling Stage 1 refits, frozen Stage 0, and matched computational budgets so architecture is not confounded with estimation policy.
3. **Make factor and state paths jointly consistent.** Condition the diffusion denoiser on predictive state-path embeddings or learn a joint discrete/continuous transition without using realized future states.
4. **Integrate one tail method at a time.** Start with training-only POT/GPD or sequential VaR recalibration, preregister its primary score, and retain the observed cross-metric trade-off.
5. **Calibrate the Wasserstein radius on a nested validation design.** If the radius remains inactive under the 20% cap, report that as a structural property and test alternative ground norms or support sets.
6. **Keep real causal work separate.** Extend the known-SCM comparators first. Open an observed-market study only with a defensible identification design.
7. **Do not evaluate post-2019 rows yet.** The diffusion experiment is underpowered and several registered ablations remain unrun.

Further research questions

- Does a simpler regime-aware block bootstrap capture most of the value attributed to the switching-factor architecture?
- Is the Stage 1 joint VaR-ES point score stable when update policies are matched?
- Which part of the observation mapping drives co-movement: B_z or residual covariance?
- Can joint factor/state diffusion improve decisions without degrading energy and variogram scores?
- At what position cap does the Wasserstein penalty begin to change allocations?
- How much structural uncertainty must a counterfactual report expose before its decision recommendation changes?

Reproducibility ledger

The Python package, YAML configurations, tests, experiment registry, receipts, artifacts, and figure-generation script are stored in the repository. The exact run order is documented in `README.md`; estimator equations and scope boundaries are in `docs/mathematical_formulation.md`; source fields, availability lags, licenses, and paid alternatives are in `docs/data_dictionary.md`; and verified research sources are in `docs/literature_review.md`.

The final release manifest is designed to bind, after all source-facing artifacts are frozen:

- the preserved Phase 0 experimental manifest;

- every completed experiment receipt and declared output;
- canonical configurations;
- final quality-assurance receipts;
- report, figures, slides, and LinkedIn material; and
- the source commit and lockfile.

The principal codebase is Python. All report figures are generated by `scripts/make_figures.py` and retained in both PNG and SVG formats.

Conclusion

CrisisForge is strongest as a dissertation-style negative-results study about evidence discipline in generative financial risk. It demonstrates a sophisticated, structured pipeline while showing that conventional time-dependence methods can remain superior on selected validation metrics, that an attractive scenario generator may not improve observed decisions, and that correct counterfactual software can still be fragile to structural assumptions.

That conclusion is more defensible-and more useful-than claiming that generative AI can simulate and survive a crisis that has never happened.

References

Primary and archival sources cited in the report. Publication status and claim boundaries are documented in docs/literature_review.md. URLs were verified for the 28 July 2026 public-core release.

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Release note

This PDF is a presentation layer over the registered Markdown report. The Python source, exact configurations, experiment receipts, lockfile, and machine-readable artifacts remain the authoritative reproducibility record. No post-2019 model evaluation is introduced by this document.